

Question 1

Consider the daily log returns of the stocks of Motorola (denoted $r_{m,t}$) and Citigroup (denoted as $r_{c,t}$) from 10 May 2017 to 1 May 2025. Suppose that Manager A holds a financial position of \$1 million dollars on each of the two stocks. Manager A decides to use RiskMetrics to calculate 5% VaR of her financial position.

Let F_{t-1} denote information set available at time $t - 1$. Under RiskMetrics, we have

$$r_{i,t}|F_{t-1} \sim N(0, \sigma_{i,t}^2)$$

where

$$\sigma_{i,t}^2 = (1 - \beta) r_{i,t-1}^2 + \beta \sigma_{i,t-1}^2$$

for $i = m, c$.

The fitted conditional variance model for Motorola is

$$\hat{\sigma}_{m,t}^2 = (1 - 0.96) r_{m,t-1}^2 + 0.96 \hat{\sigma}_{m,t-1}^2$$

and for Citigroup is

$$\hat{\sigma}_{c,t}^2 = (1 - 0.94) r_{c,t-1}^2 + 0.94 \hat{\sigma}_{c,t-1}^2$$

Table 1 provides the values of $r_{m,t}$ and $r_{c,t}$ (not expressed as percentages) along with their respective conditional variances for the last observation. Use Table 1 to answer the following questions.

- a. Calculate the 5% VaR for Motorola over the next trading day.
- b. Calculate the 5% VaR for Citigroup over the next trading day.
- c. The portfolio consists of ω_m and ω_c amounts invested in Motorola and Citigroup, respectively. Let r_{pt} denote the log return of the portfolio. Then

$$r_{pt} = \omega r_{m,t} + (1 - \omega) r_{c,t}$$

where

$$\omega = \frac{\omega_m}{\omega_m + \omega_c} \quad \text{and} \quad 1 - \omega = \frac{\omega_c}{\omega_m + \omega_c}$$

Under the RiskMetrics framework, we have

$$r_{pt}|F_{t-1} \sim N(0, \sigma_{p,t}^2).$$

The correlation between the two log returns is 0.411. What is the 5% Value-at-Risk (VaR) for the portfolio over the next trading day?

- d. Manager A has asked you to calculate the 5% VaR for her portfolio over the next trading day using the following formula:

$$\sqrt{(\text{VaR}_{0.05}^m)^2 + (\text{VaR}_{0.05}^c)^2 + 2\rho \text{VaR}_{0.05}^m \text{VaR}_{0.05}^c}$$

where $\text{VaR}_{0.05}^m$ is the 5% VaR for Motorola in part (a), $\text{VaR}_{0.05}^c$ is the 5% VaR for Citigroup in part (b), and ρ is the correlation between the two log returns.

e. Compare your results from parts (c) and (d). Are they identical? Justify your answer.

Table 1

	$r_{m,t}$	$r_{c,t}$	Fitted conditional variance of Motorola returns	Fitted conditional variance of Citigroup returns
1 May 2025	0.0629	-0.0133	0.00303	0.0103

Question 2

Consider the daily log returns r_t (*rtn.xts*) of IBM stock from 3 July 1962 to 31 December 1998. Using Figure 1, write down the fitted variance model. Determine if the leverage effect is significant at the 5% significance level and provide an explanation.

```
> spec_gjr <- ugarchspec(
+   variance.model = list(
+     model = "gjrGARCH",
+     garchorder = c(1, 1)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
> fit_gjr <- ugarchfit(
+   spec = spec_gjr,
+   data = rtn.xts*100
+ )
> fit_gjr@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.05079780	0.01347394	3.770078	1.631968e-04
omega	0.02445612	0.01367340	1.788591	7.368067e-02
alpha1	0.03166446	0.01180067	2.683275	7.290492e-03
beta1	0.93266108	0.02245369	41.537090	0.000000e+00
gamma1	0.05706979	0.01405985	4.059060	4.927066e-05

Figure 1

For fun, Difference between yields on 10-year and 2-year US bonds signals a recession (lower economic growth).

$$r_{10yr,T} - r_{2yr,T} < 0$$

